Raising a numerical inequality to a power

Squaring, cubing and taking roots — safe when both sides are positive, and full of traps when they are not.

LESSON 44–46 | 3 × 40 MIN | ALGEBRA 8, §14 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Raise both sides of an inequality to a natural power correctly.
- Take roots of both sides of an inequality.
- Use these rules to estimate the range of a square or a root.
- Give counter-examples when the positivity condition is dropped.

TEXTBOOK

National	§14, pp. 80–84
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The two rules

FOR POSITIVE NUMBERS

If $a > b > 0$ and n is a natural number, then

$$a^n > b^n \quad \text{and} \quad \sqrt[n]{a} > \sqrt[n]{b}$$

Both operations preserve the order — as long as both sides are positive. Squaring $5 > 3$ gives $25 > 9$ ✓; taking square roots of $16 > 9$ gives $4 > 3$ ✓.

What goes wrong without the condition

AN EVEN POWER DESTROYS THE ORDER

$-2 > -5$ is true, but $(-2)^2 = 4$ and $(-5)^2 = 25$, so $4 < 25$. Squaring turned the inequality round.

The reason is that an even power throws away the sign. An **odd** power keeps it, so cubing is safe for all numbers: $-2 > -5$ gives $-8 > -125$ ✓.

OPERATION	SAFE WHEN	EXAMPLE OF FAILURE
square, 4th power…	both sides positive	$-2 > -5$ but $4 < 25$
cube, 5th power…	always	—
square root	both sides ≥ 0	root of a negative does not exist

Using the rules to estimate

Given $3 < a < 5$, all positive, so squaring is safe:

$$9 < a^2 < 25$$

But given $-3 < a < 2$ you cannot square the ends. Think about what a^2 actually does on that interval: it is 0 at $a = 0$ and largest at the end furthest from 0, which is -3. So $0 \le a^2 < 9$.

THE HABIT TO BUILD

Before squaring an interval, ask whether it contains 0. If it does, the smallest value of a^2 is 0 — not the square of either end.

02 Worked examples

EXAMPLE 1 *Given $2 < a < 6$, find the range of a^2*

1. Both ends are positive, so squaring keeps the order.
   Check this first.
2. $2^2 = 4$ and $6^2 = 36$
3. $4 < a^2 < 36$

ANSWER

$4 < a^2 < 36$

EXAMPLE 2 *Given $-4 < a < 3$, find the range of a^2*

1. The interval contains 0, so the ends may not simply be squared.
2. $a^2 = 0$ at $a = 0$
   That is the smallest value.
3. The end furthest from 0 is -4, giving 16.
   That is the largest — but not attained.
4. $0 \le a^2 < 16$
   Note the ≤ on the left.

ANSWER

$0 \le a^2 < 16$

EXAMPLE 3 ***Given $4 < a < 9$, find the range of $\sqrt{a}$***

1. Both ends positive, so the root keeps the order.

2. $\sqrt{4} = 2$ and $\sqrt{9} = 3$

3. $2 < \sqrt{a} < 3$

ANSWER

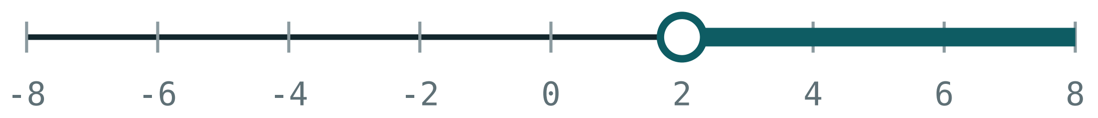
$2 < \sqrt{a} < 3$

03 Interactive model

Set the boundary either side of zero and ask what happens to the square of the interval.

Does the interval contain zero?

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Raise to a power	Darajaga ko‘tarish	Возведение в степень
Take the root	Ildiz chiqarish	Извлечь корень
Even power	Juft daraja	Чётная степень
Odd power	Toq daraja	Нечётная степень
Positive numbers	Musbat sonlar	Положительные числа
Condition	Shart	Условие
Counter-example	Qarshi misol	Контрпример
Monotonic	Monoton	Монотонный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $a > b > 0$ it follows that:

- $a^2 > b^2$
- $a^2 < b^2$
- $a^2 = b^2$
- nothing

$-2 > -5$. Squaring gives:

- $4 > 25$
- $4 < 25$
- $4 = 25$
- the same inequality

Cubing an inequality is safe:

- only for positives
- for all numbers
- never
- only for whole numbers

If $-3 < a < 2$ then a^2 satisfies:

- $9 < a^2 < 4$
- $0 \leq a^2 < 9$
- $4 < a^2 < 9$
- $-9 < a^2 < 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *$5 > 3$. Square both sides.*

   ANSWER $25 > 9$

2. *$7 > 2$. Cube both sides.*

   ANSWER $343 > 8$

3. *$16 > 9$. Take square roots.*

   ANSWER $4 > 3$

4. *$1 < a < 4$. Find the range of a^2*

   ANSWER $1 < a^2 < 16$

5. *$9 < a < 25$. Find the range of $\sqrt{a}$*

   ANSWER $3 < \sqrt{a} < 5$

6. *Is squaring safe for $a > b > 0$?*

   ANSWER Yes.

7. *Is cubing safe for negative numbers?*

   ANSWER Yes — an odd power keeps the sign.

Medium · the standard question

7 PROBLEMS

1. $2 < a < 6$. *Find the range of* a^2

   ANSWER $4 < a^2 < 36$

2. $4 < a < 9$. *Find the range of* $\sqrt{a}$

   ANSWER $2 < \sqrt{a} < 3$

3. $1 < a < 2$. *Find the range of* a^3

   ANSWER $1 < a^3 < 8$

4. $-5 < a < -2$. *Find the range of* a^2

   ANSWER $4 < a^2 < 25$

5. $-5 < a < -2$. *Find the range of* a^3

   ANSWER $-125 < a^3 < -8$

6. *Show that* $-2 > -5$ *but* $(-2)^2 < (-5)^2$

   ANSWER $4 < 25$ — squaring reversed it.

7. $3 < a < 4$. *Find the range of* $\frac{1}{a^2}$

   ANSWER $\frac{1}{16} < \frac{1}{a^2} < \frac{1}{9}$

Hard · stretch and reason

7 PROBLEMS

1. $-4 < a < 3$. Find the range of a^2

ANSWER $0 \le a^2 < 16$

2. $-1 < a < 5$. Find the range of a^2

ANSWER $0 \le a^2 < 25$

3. $2 < a < 3$. Find the range of $a^2 + a$

ANSWER $4 < a^2 < 9$, so $6 < a^2 + a < 12$

4. The side of a square satisfies $4 < a < 6$. Estimate its area and perimeter.

ANSWER $16 < S < 36$, $16 < P < 24$

5. Compare $\sqrt{5}$ and 2.3 without a calculator.

ANSWER $2.3^2 = 5.29 > 5$, so $\sqrt{5} < 2.3$

6. $0 < a < 1$. Compare a^3, a^2 and a.

ANSWER $a^3 < a^2 < a$ — each multiplication by $a < 1$ makes it smaller.

7. Given $1 < a < 4$, find the range of $\sqrt{a} + a$

ANSWER $1 < \sqrt{a} < 2$, so $2 < \sqrt{a} + a < 6$

07 Homework

Homework — 6 problems

Algebra 8, §14, pp. 80–84. Check for zero inside the interval before squaring.

1. $3 < a < 7$. Find the range of a^2
2. $1 < a < 8$. Find the range of $\sqrt[3]{a}$
3. $-6 < a < 2$. Find the range of a^2
4. $-3 < a < -1$. Find the range of a^3
5. Compare $\sqrt{7}$ and 2.6 without a calculator

6. *The side of a square satisfies $5 < a < 8$. Estimate its area.*